

Ratios & Proportions

Using Physical examples

Key topics

Terminology and notation

Conversion between ratios, proportions and percentages

Calculating mixtures of solutions and compounds

Introduction to simultaneous equations

Terminology

A ratio is a comparison between two similar quantities

If 72 cm of a full sized object is represented by 1cm in a model of that object then the scale ratio is 1 to 72

If the model is **a** cm long and the full sized object is **b** cm long then a is to b as 1 is to 72

We write $a:b$ $::$ $1:72$ Say “a to b is defined as one to seventy two”

Or $\frac{a}{b} = \frac{1}{72}$

Or $72a=b$

Or $a = \frac{b}{72}$

Example 1

Two speeds, $U\text{ms}^{-1}$ and $V\text{ms}^{-1}$ are in the ratio 8:3 what is the value of V?

If U is 5ms^{-1} , what is the value of V?

We are given that $U:V :: 8:3$ or $\frac{U}{V} = \frac{8}{3}$

We are told that $U=5$ so $\frac{5}{V} = \frac{8}{3}$

Cross multiplying gives $15=8V$, so $V = \frac{15}{8} = 1.875$

Example 2a

2.504g of Copper Oxide contains 2g of Copper. How much Copper is contained in 0.939g of Copper Oxide?

Assume the answer is x g, then $2:x :: 2.504:0.939$

$$\frac{2}{x} = \frac{2.504}{0.939}$$

Cross multiplying gives $2 \times 0.939 = 2.504x$

$$x = \frac{2 \times 0.939}{2.504} = 0.75g$$

Example 2b

2.504g of Copper Oxide contains 2g of Copper. How much Copper is contained in 0.939g of Copper Oxide?

2.504g of Copper Oxide contains 2g of Copper

Therefore 1g of Copper oxide contains $\frac{2}{2.504}$ g of Copper

Therefore 0.939g of Copper Oxide contains $0.939 \times \frac{2}{2.504} = 0.75$ g of copper

Example 3

A quantity of alloy with a mass of 336g contains Copper, Lead and Tin in the ratio 16:4:1 by mass. To this alloy are added 4g of Copper, 36g of Lead and 4g of Tin. Find the ratios, by mass of the Copper, Lead and Tin in the new alloy.

Originally, we have $16:4:1 = \text{21 equal parts}$ which means that each part is equal to $\frac{336}{21} = 16\text{g}$

Giving: $16 \times 16 = 256\text{g}$ of Copper

$4 \times 16 = 64\text{g}$ of Lead

$1 \times 16 = 16\text{g}$ of Tin

In the new alloy, there are $(256 + 4)\text{g} = 260\text{g}$ of Copper
 $(64 + 36)\text{g} = 100\text{g}$ of Lead
 $(16 + 4)\text{g} = 20\text{g}$ of Tin

Making 380g for the new alloy

So Copper:Lead:Tin :: 260:100:20

Or Copper:Lead:Tin :: 13:5:1

Tricks for non integer answers

In a lot of questions I pick numbers to make the answers simple integers, where that isn't the case, here is a simple trick to help

Multiply by 10 for single decimal place, 100 for two etc.

Divide by the smallest number

Multiply by 100

e.g 262:96.7:18.1

Multiply by 10 2620:967:181

Divide by 181 14.48:5.34:1

Multiply by 100 1448:534:100

Percentages

Awkward numbers in ratios make the idea of percentages more attractive.

Whatever the whole happens to be it is considered to consist of 100 equal parts.

$$5\% = \frac{5}{100} = 0.05$$

$$125\% = \frac{125}{100} = 1.25$$

$$0.75 = \frac{75}{100} = 75\%$$

$$1.583 = \frac{1.583}{100} = 158.3\%$$

Guide

$x\%$ of y is found by evaluating $\frac{x}{100} \times y$

$$15\% \text{ of } 23 \text{ is } \frac{15}{100} \times 23 = 0.15 \times 23 = 3.45$$

$$125\% \text{ of } 20 \text{ is } \frac{125}{100} \times 20 = 1.25 \times 20 = 25$$

25% of mass A is 14kg, what is the mass of A?

$$1\% \text{ of Mass A is } \frac{14}{25} \text{kg,}$$

$$\text{So } 100\% \text{ of mass A is } 100 \times \frac{14}{25} = 56 \text{kg}$$

£1 example

Percentage change 1

A Volume of 56.0ml must be increased by 10%, What is the new volume?

Old volume 56ml = 100%

New volume v ml = 110%

Therefore $\frac{v}{56} = \frac{110}{100}$

$$v = 56 \times \frac{110}{100} = 61.6$$

Percentage change 2

A Volume of 61.6ml must be decreased by 10%, What is the new volume?

Old volume 61.6ml = 100%

New volume v ml = 90%

Therefore $\frac{v}{61.6} = \frac{90}{100}$ $v = 61.6 \times \frac{90}{100} = 55.4$

Evaluating mixtures

250mg of A mixed with B make a mixture of mass 1.21g, what percentage of B is there in the mixture?

Let x be the amount of B in mg.

$$250\text{mg} + B\text{mg} = 1.21\text{g} = 1210\text{mg}$$

$$B = 1210 - 250 = 960$$

The percentage of B in the mixture is therefore $\frac{960}{1210} \times 100 = 79.3\%$

Reality check

The Mathematics for mixing solutions is fairly clear, but it only applies to ideal volumes of liquids which are very chemically similar.

Mix 100ml of pure alcohol with 100ml of water and how much solution do you have?

190ml

Yes, I know – Chemistry stinks.

Mixing solutions 1

50ml of 20% solution is added to 100ml of 10% solution, both being mixtures of liquid A in water. What is the strength of the resulting solution?

The 50ml of the 20% solution contains $20\% \times 50\text{ml} = 10\text{ml of A}$ and $80\% \times 50\text{ml} = 40\text{ml of Water}$

The 100ml of the 10% solution contains $10\% \times 100\text{ml} = 10\text{ml of A}$ and $90\% \times 100\text{ml} = 90\text{ml of Water}$

The new solution therefore contains $(10+10)\text{ml of A}$ and $(40+90)\text{ml of Water}$, with a total volume of 150ml

Percentage strength of the solution is therefore $\frac{20}{150} \times 100 = 13.\dot{3}\%$

Mixing solutions 2

How much 15% solution of A needs to be added to 60ml of 25% solution to achieve a 17.5% solution?

Let V be the volume of our 15% solution that we need to add.

The volume of the new solution is $60+V$ ml

The amount of A in the 25% solution is 25% of 60ml=15ml

The amount of A in the 15% solution is $0.15V$

The amount of A in the new solution is $15+0.15V$ ml

The strength of the new solution is given by $\frac{15+0.15V}{60+V} \times 100 = 17.5$

Mixing solutions 3

$$\frac{15 + 0.15V}{60 + V} \times 100 = 17.5$$

Multiplying denominator: $(15 + 0.15V) \times 100 = 17.5(60 + V)$

Expanding: $1500 + 15V = 1050 + 17.5V$

Rearranging: $1500 - 1050 = 17.5V - 15V$
 $450 = 2.5V$

$$V = \frac{450}{2.5} = 180$$

So 180ml of the 15% solution should be added

Compounds 1

S_1 : Solution 1 has 80g of C in 1 litre (or 0.08g in 1ml)

S_2 : Solution 2 has 25g of C in 1 litre (or 0.025g in 1ml)

How much of each solution is needed to make 1 litre of 60g/litre solution?

We need x ml of S_1 and y ml of S_2 to make S_3

The total amount of compound C is $0.08x + 0.025y = 60$ g

Total volume is $x + y = 1000$ ml

Compounds 2:

(1)

$$x+y=1000$$

(2)

$$0.08x+0.025y=60$$

Two equations and two unknowns (x, y)

(3) $0.08 \times (1)$

$$0.08x+0.08y=80$$

(3) Rearranged

$$0.08x = 80 - 0.08y$$

(4) Substitute (3) into (2)

$$80 - 0.08y + 0.025y = 60$$

(4) Rearranged

$$0.055y = 20$$

(4) Solved

$$y = \frac{20}{0.55} = 364$$

(1) Solved

$$x = 1000 - y = 636$$

Compounds 3: How much of each solution is needed to make 70ml of 45g/litre solution?

First, we note that 70ml of 45g/l means $\frac{70}{1000} \times 45 = 3.15g$

$$x+y=70 \quad (1)$$

$$0.08x+0.025y = 3.15 \quad (2)$$

$$(1) \times 0.08 \quad 0.08x+0.08y=5.6 \quad (3)$$

$$(2)-(3) \quad -0.055y = -2.45 \quad (4)$$

$$\text{Rearranging (4)} \quad y = \frac{2.45}{0.055} = \mathbf{44.5} \quad (5)$$

$$\text{Substituting } y \text{ in (1)} \quad x + 44.5 = 70 \quad (6)$$

$$\text{Rearranging (6)} \quad x = 70 - 44.5 = \mathbf{25.5} \quad (7)$$

Simultaneous equations

We could only solve the problem because we had TWO equations.

Each related the same two variables but both were **independent**.

One equation related the masses of the liquids, one related the concentrations

Try solving these:

$$x + 2y = 10$$

$$3x + 6y = 30$$

Independence

- | | |
|--------------------------|------------------------|
| (1) | $x + 2y = 10$ |
| (2) | $3x + 6y = 30$ |
| (3) Rearranging (1) | $x = 10 - 2y$ |
| (4) Subbing (3) into (2) | $3(10 - 2y) + 6y = 30$ |
| (5) Rearranging (4) | $30 - 6y + 6y = 30$ |

In this case, each equation contained the same information, so the techniques failed – we didn't REALLY have two equations, just one.

We say the equations were not **independent**.

Advice for dealing with problems

Start by gathering info from the question

Simple questions often need you to 'count the parts' of different ingredients

Consider if you can make an equation to describe the situation

If you can make two separate equations then you might need two, simultaneous equations...